ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2024–2025**

DURATION: 2 HOURS **FULL MARKS: 120**

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

---

1. Jordan, a passionate sneakerhead, wants to crack the code behind the pricing of trendy new sneakers. He suspects that the price of a sneaker depends on two main factors: (a) the comfort rating, which measures how comfortable the shoe feels during wear, and (b) the hype score, which reflects how popular and sought-after the sneaker is in the sneakerhead community. To test his theory, Jordan collected data on four of the latest sneaker releases shown in Table 1.

Table 1: A dataset of sneaker prices based on comfort rating and hype score.

Sneaker	Comfort Rating (x_1)	Hype Score (x_2)	Price (USD) (y)
Nike Air Force 1	8	7	180
Adidas Superstar	6	9	210
Converse	7	6	165
Vans Old Skool	9	8	220

Answer the following questions based on the dataset in Table 1 assuming that the price y depends linearly on the comfort rating and hype score, i.e.,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

a) Formulate the problem of finding the best-fitting line in $Ax = b$ form where A, b, and x are the coefficient matrix, constant vector, and variable vector respectively, and determine the following: 12 (CO3) (PO1)

i. The coefficients of the best fit line (β_0, β_1, and β_2) as $\hat{x}$.

ii. The projection of all data (p) onto the column space of A.

iii. The projection matrix P that projects b onto the column space of A.

iv. The error vector e and the mean squared error (MSE) of the fit.

Solution:

We model the price y as a linear function of comfort rating x_1 and hype score x_2:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

For the four sneakers, the matrix A (including a column of ones for the intercept) and the vector b of observed prices are:

$$A = \begin{bmatrix} 1 & 8 & 7 \\ 1 & 6 & 9 \\ 1 & 7 & 6 \\ 1 & 9 & 8 \end{bmatrix}, \qquad b = \begin{bmatrix} 180 \\ 210 \\ 165 \\ 220 \end{bmatrix}.$$

The normal equations are $A^\top A\,\hat{x} = A^\top b$ with $\hat{x} = [\beta_0, \beta_1, \beta_2]^\top$.

i) Coefficients $\hat{x}$

$$A^\top A = \begin{bmatrix} 4 & 30 & 30 \\ 30 & 230 & 224 \\ 30 & 224 & 230 \end{bmatrix}, \qquad A^\top b = \begin{bmatrix} 775 \\ 5835 \\ 5900 \end{bmatrix}.$$

Solving the system yields,

$$\hat{x} = (A^T A)^{-1}(A^T b)$$

$$\boxed{\hat{x} = \begin{bmatrix} -\frac{25}{2} \\ \frac{25}{3} \\ \frac{115}{6} \end{bmatrix}}.$$

Thus $\beta_0 = -\frac{25}{2}$ or -12.5, $\beta_1 = \frac{25}{3}$ or $8.\overline{3}$, $\beta_2 = \frac{115}{6}$ or $19.1\overline{6}$.

ii) Projection $p = A\hat{x}$

$$p = A\hat{x} = \begin{bmatrix} 1 & 8 & 7 \\ 1 & 6 & 9 \\ 1 & 7 & 6 \\ 1 & 9 & 8 \end{bmatrix} \begin{bmatrix} -25/2 \\ 25/3 \\ 115/6 \end{bmatrix} = \begin{bmatrix} \frac{565}{3} \\ 210 \\ \frac{965}{6} \\ \frac{1295}{6} \end{bmatrix} \approx \begin{bmatrix} 188.333 \\ 210 \\ 160.833 \\ 215.833 \end{bmatrix}.$$

Thus,

$$\boxed{p = \begin{bmatrix} -25/2 \\ 25/3 \\ 115/6 \end{bmatrix}}$$

iii) Projection matrix $P = A(A^\top A)^{-1}A^\top$

First compute $(A^\top A)^{-1}$:

$$(A^\top A)^{-1} = \frac{1}{96} \begin{bmatrix} 2724 & -180 & -180 \\ -180 & 20 & 4 \\ -180 & 4 & 20 \end{bmatrix} = \begin{bmatrix} \frac{227}{8} & -\frac{15}{8} & -\frac{15}{8} \\ -\frac{15}{8} & \frac{5}{24} & \frac{1}{24} \\ -\frac{15}{8} & \frac{1}{24} & \frac{5}{24} \end{bmatrix}.$$

Then $P = A(A^\top A)^{-1}A^\top$ simplifies to

$$\boxed{P = \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 1 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{5}{6} & -\frac{1}{6} \\ \frac{1}{3} & 0 & -\frac{1}{6} & \frac{5}{6} \end{bmatrix}}.$$

4. Error vector e and Mean Squared Error (MSE)

$$e = b - p = \begin{bmatrix} 180 \\ 210 \\ 165 \\ 220 \end{bmatrix} - \begin{bmatrix} 565/3 \\ 210 \\ 965/6 \\ 1295/6 \end{bmatrix} = \begin{bmatrix} -\frac{25}{3} \\ 0 \\ \frac{25}{6} \\ \frac{25}{6} \end{bmatrix}.$$

$$\text{MSE} = \frac{1}{4} e^{\top} e = \frac{1}{4}\left(\frac{625}{9} + 0 + \frac{625}{36} + \frac{625}{36}\right) = \frac{1}{4} \cdot \frac{625}{6} = \boxed{\frac{625}{24}}.$$

Rubric:

- 4 points for correctly determining $\hat{x}$.
- 2 points for correctly determining $\hat{p}$.
- 2 points for correctly determining $\hat{P}$.
- 4 point for correctly determining e and MSE.

b) An outlier is a data point that significantly deviates from the overall pattern of the data. Curious to verify the results obtained in Question 1.a, Jordan decided to expand his dataset by collecting information on a few more sneakers. While reviewing the data of the new sneakers, he noticed that the price of one sneaker seemed unusually different from what his regression model would predict, which he suspects to be an outlier. The new dataset is shown in Table 2. Identify which data point is most likely the outlier and justify your answer. Your conclusion must be supported by appropriate calculations based on results obtained in Question 1.a. 8 (CO3) (PO1)

Table 2: A dataset of prices of new sneakers.

Sneaker	Comfort Rating (x_1)	Hype Score (x_2)	Price (USD) (y)
New Balance 990	9	6	175
Nike Air Jordan 1	8	12	295
Abidas UltraBoost	5	9	360
Nike Dunk	7	9	225

Solution:

We use the regression coefficients $\hat{x} = \begin{bmatrix} -\frac{25}{2} \\ \frac{25}{3} \\ \frac{115}{6} \end{bmatrix}$ obtained in Question 1.a. For the new sneakers, we form the matrix A_{new} (including intercept column) and the observed price vector b_{new}:

$$A_{\text{new}} = \begin{bmatrix} 1 & 9 & 6 \\ 1 & 8 & 12 \\ 1 & 5 & 9 \\ 1 & 7 & 9 \end{bmatrix}, \qquad b_{\text{new}} = \begin{bmatrix} 175 \\ 295 \\ 360 \\ 225 \end{bmatrix}.$$

The predicted prices are $\hat{y}_{\text{new}} = A_{\text{new}}\hat{x}$:

$$A_{\text{new}}\hat{x} = \begin{bmatrix} 1 & 9 & 6 \\ 1 & 8 & 12 \\ 1 & 5 & 9 \\ 1 & 7 & 9 \end{bmatrix} \begin{bmatrix} -\frac{25}{2} \\ \frac{25}{3} \\ \frac{115}{6} \end{bmatrix} = \begin{bmatrix} -\frac{25}{2} + 9 \cdot \frac{25}{3} + 6 \cdot \frac{115}{6} \\ -\frac{25}{2} + 8 \cdot \frac{25}{3} + 12 \cdot \frac{115}{6} \\ -\frac{25}{2} + 5 \cdot \frac{25}{3} + 9 \cdot \frac{115}{6} \\ -\frac{25}{2} + 7 \cdot \frac{25}{3} + 9 \cdot \frac{115}{6} \end{bmatrix} = \begin{bmatrix} -\frac{25}{2} + 75 + 115 \\ -\frac{25}{2} + \frac{200}{3} + 230 \\ -\frac{25}{2} + \frac{125}{3} + 172.5 \\ -\frac{25}{2} + \frac{175}{3} + 172.5 \end{bmatrix} = \begin{bmatrix} 177.5 \\ 284.1\overline{6} \\ 201.6\overline{6} \\ 218.3\overline{3} \end{bmatrix}.$$

Now compute the residual vector $e_{\text{new}} = b_{\text{new}} - \hat{y}_{\text{new}}$:

$$e_{\text{new}} = \begin{bmatrix} 175 \\ 295 \\ 360 \\ 225 \end{bmatrix} - \begin{bmatrix} 177.5 \\ 284.1\overline{6} \\ 201.6\overline{6} \\ 218.3\overline{3} \end{bmatrix} = \begin{bmatrix} -2.5 \\ 10.833\ldots \\ 158.333\ldots \\ 6.666\ldots \end{bmatrix}.$$

The third residual (Adidas UltraBoost) is 158.33, far larger in magnitude than the others (all below 11). Therefore, the outlier is the **Adidas UltraBoost**.

Rubric:

- 4 points for correctly determining the predicted prices.
- 2 points for computing the difference or any step to detect outlier.
- 2 points for identifying the outlier and justification.

c) Do you think the regression model should depend on both the comfort rating and the hype score, or would a single factor be sufficient? If a single factor is sufficient, determine which one provides the better model. Justify your answer mathematically by comparing the fitting errors using the data given in Table 1. 10 (CO3) (PO1)

Assume that a regression model using a single factor has the form

$$y = \beta_0 + \beta_1 x_i + \beta_2 x_i^2,$$

where x_i represents either the comfort rating or the hype score. Compute the corresponding errors and use them to support your conclusion. Do not include the data from Table 2 in your calculations.

Solution:

We compare the mean squared errors (MSE) of three models using the data from Table 1:

Sneaker	x_1	x_2	y
Nike Air Force 1	8	7	180
Adidas Superstar	6	9	210
Converse	7	6	165
Vans Old Skool	9	8	220

Model 1: Two variable linear ($y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$)

From Question 1.a, the leastsquares fit gives

$$\hat{y} = -\frac{25}{2} + \frac{25}{3}x_1 + \frac{115}{6}x_2,$$

with error vector $e = \begin{bmatrix} -25/3 \\ 0 \\ 25/6 \\ 25/6 \end{bmatrix}$.

Sum of squared errors:

$$\text{SSE} = \left(\frac{25}{3}\right)^2 + 0^2 + \left(\frac{25}{6}\right)^2 + \left(\frac{25}{6}\right)^2 = \frac{625}{6}.$$

$$\text{MSE}_{\text{both}} = \frac{\text{SSE}}{4} = \frac{625}{24} \approx 26.0417.$$

Model 2: Quadratic in comfort rating ($y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2$)

Matrix $A = \begin{bmatrix} 1 & 8 & 64 \\ 1 & 6 & 36 \\ 1 & 7 & 49 \\ 1 & 9 & 81 \end{bmatrix}$, $b = \begin{bmatrix} 180 \\ 210 \\ 165 \\ 220 \end{bmatrix}$. Solving the normal equations yields

$$\hat{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \frac{5315}{4} \\ -\frac{1257}{4} \\ \frac{85}{4} \end{bmatrix} \Rightarrow \hat{y} = \frac{1}{4}(5315 - 1257x_1 + 85x_1^2).$$

Residuals: $e = \begin{bmatrix} 21/4 \\ 7/4 \\ -21/4 \\ -7/4 \end{bmatrix}$.

$$\text{SSE} = \left(\frac{21}{4}\right)^2 + \left(\frac{7}{4}\right)^2 + \left(\frac{21}{4}\right)^2 + \left(\frac{7}{4}\right)^2 = \frac{980}{16} = \frac{245}{4} = 61.25.$$

$$\text{MSE}_{\text{comfort}} = \frac{61.25}{4} = 15.3125.$$

Model 3: Quadratic in hype score ($y = \beta_0 + \beta_1 x_2 + \beta_2 x_2^2$)

Matrix matrix $A = \begin{bmatrix} 1 & 7 & 49 \\ 1 & 9 & 81 \\ 1 & 6 & 36 \\ 1 & 8 & 64 \end{bmatrix}$, same b. The normal equations give

$$\hat{x} = \begin{bmatrix} -\frac{1125}{4} \\ \frac{445}{4} \\ -\frac{25}{4} \end{bmatrix} \Rightarrow \hat{y} = \frac{1}{4}(-1125 + 445x_2 - 25x_2^2).$$

Residuals: $e = \begin{bmatrix} -45/4 \\ -15/4 \\ 15/4 \\ 45/4 \end{bmatrix}$.

$$\text{SSE} = \left(\frac{45}{4}\right)^2 + \left(\frac{15}{4}\right)^2 + \left(\frac{15}{4}\right)^2 + \left(\frac{45}{4}\right)^2 = \frac{2250}{16} = \frac{1125}{8} = 140.625.$$

$$\text{MSE}_{\text{hype}} = \frac{140.625}{4} = 35.15625.$$

Comparison

$$\text{MSE}_{\text{both}} \approx 26.04, \quad \text{MSE}_{\text{comfort}} = 15.31, \quad \text{MSE}_{\text{hype}} = 35.16.$$

The quadratic model using only comfort rating has the smallest mean squared error. Therefore, **a single factor (comfort rating, with a quadratic term) is sufficient and provides a better fit than the two - variable linear model**. The hype score alone does not perform as well, neither does the combination of both.

Rubric:

- 4 points for correctly setting up and solving the quadratic comfort model.
- 4 points for correctly setting up and solving the quadratic hype model.
- 1 point for comparing MSEs and concluding that comfort alone is sufficient.
- 1 point for identifying the better single factor (comfort rating).

2. a) Find Singular Value Decomposition (SVD) of the following matrix A and express it in $A = U\Sigma V^T$ form, identifying the matrices U, Σ and V. 15 (CO3) (PO1)

$$A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$

Solution:

Given,

$$A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$$

The transpose of A (A^T) is:

$$A^T = \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix}$$

Now, $A^T A$:

$$A^T A = \begin{bmatrix} 4 & 8 \\ 11 & 7 \\ 14 & -2 \end{bmatrix} \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix} = \begin{bmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix}$$

To find the eigenvalues, we solve the characteristic equation:

$$\det(A^T A - \lambda I) = 0$$

Where λ represents the eigenvalues and I is the identity matrix.

$$A^T A - \lambda I = \begin{bmatrix} 80-\lambda & 100 & 40 \\ 100 & 170-\lambda & 140 \\ 40 & 140 & 200-\lambda \end{bmatrix}$$

The determinant is:

$$\det(A^T A - \lambda I) = 0$$

We note that $\text{rank}(A) = 2$, so one eigenvalue is 0. The other two eigenvalues are the same as those of AA^T:

$$AA^T = \begin{bmatrix} 333 & 81 \\ 81 & 117 \end{bmatrix}$$

$$\det(AA^T - \lambda I) = (333-\lambda)(117-\lambda) - 81^2 = \lambda^2 - 450\lambda + 32400 = 0$$

$$\lambda = 225 \pm 135 \quad \Rightarrow \quad \lambda_1 = 360, \quad \lambda_2 = 90$$

Thus, the eigenvalues of $A^T A$ are:

$$\lambda_1 = 360, \quad \lambda_2 = 90, \quad \lambda_3 = 0$$

Now, finding eigenvectors of $A^T A$:
For $\lambda = 360$ (descending order):

$$\begin{bmatrix} 80-360 & 100 & 40 \\ 100 & 170-360 & 140 \\ 40 & 140 & 200-360 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} -280 & 100 & 40 \\ 100 & -190 & 140 \\ 40 & 140 & -160 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

Solving yields $x_1 = t/2,\ y_1 = t,\ z_1 = t$. Choose $t = 2$:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

After normalizing (norm $= \sqrt{1+4+4} = 3$):

$$v_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

For $\lambda = 90$:

$$\begin{bmatrix} 80-90 & 100 & 40 \\ 100 & 170-90 & 140 \\ 40 & 140 & 200-90 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

Solving yields $x_1 = 2t,\ y_1 = t,\ z_1 = -2t$. Choose $t = 1$:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

After normalizing (norm $= \sqrt{4+1+4} = 3$):

$$v_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix}$$

For $\lambda = 0$:

$$\begin{bmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

Solving yields $x_1 = 2t,\ y_1 = -2t,\ z_1 = t$. Choose $t = 1$:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

After normalizing (norm $= \sqrt{4+4+1} = 3$):

$$v_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}$$

Thus,

$$V = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

Again,

$$AA^T = \begin{bmatrix} 333 & 81 \\ 81 & 117 \end{bmatrix}$$

For $\lambda = 360$:

$$\begin{bmatrix} 333 - 360 & 81 \\ 81 & 117 - 360 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0$$

$$\begin{bmatrix} -27 & 81 \\ 81 & -243 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0$$

We get $x_1 = 3y_1$. Choose $y_1 = 1$:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Normalizing (norm $= \sqrt{9+1} = \sqrt{10}$):

$$u_1 = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

For $\lambda = 90$:

$$\begin{bmatrix} 333 - 90 & 81 \\ 81 & 117 - 90 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 243 & 81 \\ 81 & 27 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0$$

We get $3x_1 + y_1 = 0 \Rightarrow y_1 = -3x_1$. Choose $x_1 = 1$:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Normalizing (norm $= \sqrt{1+9} = \sqrt{10}$):

$$u_2 = \begin{bmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix}$$

Thus,

$$U = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix}$$

The singular values are $\sigma_1 = \sqrt{360} = 6\sqrt{10}$, $\sigma_2 = \sqrt{90} = 3\sqrt{10}$.
So,

$$\Sigma = \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix}$$

Therefore,

$$A = U\Sigma V^T$$

$$\Rightarrow A = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix} \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}^T$$

Rubric:

- 1 point for correctly writing $A^T A$ and AA^T.
- 1 point each for correctly identifying eigenvalues (times 2).
- 2 points for each for correctly identifying each eigenvector (times 5).
- 1 point for correctly identifying Σ.
- 1 point for correct $A = U\Sigma V^T$ representation.

Note: Your answer for V and U might be different, but any correct answer will be accepted.

b) A sequence is given by $2 + 7 + 1$ (CO3) (PO1)

$$-1,\ 3,\ 5,\ 27,\ 101,\ 411,\ 1637,\ \cdots$$

where each term is obtained as a linear combination of the two preceding terms. The first two terms of the sequence are $a_0 = -1$ and $a_1 = 3$.

i. Determine the recurrence relation for the sequence. In other words, express a_n in terms of a_{n-1} and a_{n-2}.

Solution:

Given sequence: $a_0 = -1$, $a_1 = 3$, and subsequent terms are linear combinations of the two preceding terms. From the given values:

$$a_2 = 5, \quad a_3 = 27, \quad a_4 = 101, \quad a_5 = 411, \quad a_6 = 1637.$$

Assume $a_n = p\,a_{n-1} + q\,a_{n-2}$. Using $n = 2$ and $n = 3$:

$$\begin{cases} 5 = p \cdot 3 + q \cdot (-1) & \Rightarrow 3p - q = 5, \\ 27 = p \cdot 5 + q \cdot 3 & \Rightarrow 5p + 3q = 27. \end{cases}$$

Solving: multiply first by 3: $9p - 3q = 15$, add to second: $14p = 42 \Rightarrow p = 3$, then $q = 3p - 5 = 9 - 5 = 4$. Check with $n = 4$: $3 \cdot 27 + 4 \cdot 5 = 81 + 20 = 101$, correct. Thus the recurrence is

$$\boxed{a_n = 3a_{n-1} + 4a_{n-2}} \quad (n \geq 2).$$

Rubric:

- 1 point for formulating the two equations.
- 1 point for solving the two equations and determining the correct recurrence relation.

ii. Express the recurrence relation obtained in Question 2.b)i in Matrix form and use diagonalization to derive a formula for the k-th term of the sequence.

Solution:

We write the recurrence in matrix form. Let $\mathbf{V}_n = \begin{bmatrix} a_n \\ a_{n-1} \end{bmatrix}$ for $n \geq 1$. Then

$$\mathbf{V}_n = \begin{bmatrix} 3 & 4 \\ 1 & 0 \end{bmatrix} \mathbf{V}_{n-1}.$$

Denote $M = \begin{bmatrix} 3 & 4 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{V}_1 = \begin{bmatrix} a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. By induction,

$$\mathbf{V}_n = M^{n-1}\mathbf{V}_1 \quad (n \geq 1).$$

Diagonalization: The characteristic polynomial of M is

$$\det(M - \lambda I) = \begin{vmatrix} 3-\lambda & 4 \\ 1 & -\lambda \end{vmatrix} = (3-\lambda)(-\lambda) - 4 = \lambda^2 - 3\lambda - 4.$$

Let λ_1, λ_2 be its two roots ($\lambda_1 = 4$, $\lambda_2 = -1$ after substitution, but we keep them symbolic). For an eigenvalue λ, solve $(M - \lambda I)\mathbf{v} = 0$:

$$\begin{bmatrix} 3-\lambda & 4 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{cases} (3-\lambda)x + 4y = 0, \\ x - \lambda y = 0. \end{cases}$$

From the second equation, $x = \lambda y$. Choosing $y = 1$ gives the eigenvector $\mathbf{v} = \begin{bmatrix} \lambda \\ 1 \end{bmatrix}$.

Thus the eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}.$$

Form the matrix $S = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}$ and the diagonal matrix $\Lambda = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$.

Then $M = S\Lambda S^{-1}$ and $M^{n-1} = S\Lambda^{n-1}S^{-1}$.

Compute S^{-1}. Since $\det S = \lambda_1 \cdot 1 - \lambda_2 \cdot 1 = \lambda_1 - \lambda_2$,

$$S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}.$$

Now $\mathbf{V}_n = S\Lambda^{n-1}S^{-1}\mathbf{V}_1$.

$$\Rightarrow \mathbf{V}_n = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}^{n-1} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\Rightarrow \mathbf{V}_n = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^{n-1} & 0 \\ 0 & \lambda_2^{n-1} \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$\Rightarrow \mathbf{V}_n = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^{n-1} & 0 \\ 0 & \lambda_2^{n-1} \end{bmatrix} \begin{bmatrix} 3 + \lambda_2 \\ -3 - \lambda_1 \end{bmatrix}$$

$$\Rightarrow \mathbf{V}_n = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^{n-1}(3 + \lambda_2) \\ -\lambda_2^{n-1}(3 + \lambda_1) \end{bmatrix}$$

$$\Rightarrow \mathbf{V}_n = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1^{n}(3 + \lambda_2) - \lambda_2^{n}(3 + \lambda_1) \\ \lambda_1^{n-1}(3 + \lambda_2) - \lambda_2^{n-1}(3 + \lambda_1) \end{bmatrix}$$

Thus a_n in terms of the eigenvalues λ_1, λ_2 is

$$\boxed{a_n = \frac{\lambda_1^n(3+\lambda_2) - \lambda_2^n(3+\lambda_1)}{\lambda_1 - \lambda_2}} \qquad (n \geq 1),$$

with $a_0 = -1$. For the actual eigenvalues $\lambda_1 = 4,\ \lambda_2 = -1$, substitute:

$$3 + \lambda_2 = 2, \quad 3 + \lambda_1 = 7, \quad \lambda_1 - \lambda_2 = 5,$$

giving

$$a_n = \frac{4^n \cdot 2 - (-1)^n \cdot 7}{5} = \frac{2 \cdot 4^n - 7(-1)^n}{5}.$$

Rubric:

- 1 point for correctly writing the matrix form.
- 1 points for correct diagonalization of the Matrix.
- 4 points for correctly deriving the kth term (or n^{th} term).
- 1 point for correctly deriving a_n.

iii. Using the result obtained in Question 2.b)ii, compute the 10th term of the sequence.

Solution:

From Question 2.b)ii we obtained $a_n = \frac{2 \cdot 4^n - 7(-1)^n}{5}$.

So, for $n = 10$, $a_{10} = \frac{2 \cdot 4^{10} - 7(-1)^{10}}{5}. = 419429$

Rubric:

- 1 point for showing $a_{10} = 419429$.

c) Given, the bases of Columnspace, $C(D)$ and Left Nullspace, $N(D^T)$ of matrix D in $\mathbb{R}^5$. Determine the values of p,q and r. 5 (CO3) (PO1)

Table 3: Bases of $C(A)$ and $N(A^T)$

Columnspace, $C(D)$	Left Nullspace, $N(D^T)$
$\begin{bmatrix} 1 \\ 2 \\ p \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} q \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix}, \begin{bmatrix} -2 \\ -2 \\ -4 \\ -3 \\ 0 \end{bmatrix}$	$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ 4 \\ 0 \\ r \\ 1 \end{bmatrix}$

Solution:

Given the bases of the columnspace $C(D)$ and the left nullspace $N(D^T)$ of matrix D in $\mathbb{R}^5$. Since $C(D)$ and $N(D^T)$ are orthogonal complements in $\mathbb{R}^5$, every vector in $C(D)$ is orthogonal to every vector in $N(D^T)$. Thus, the dot product of each basis vector of $C(D)$ with each basis vector of $N(D^T)$ must be zero.

Let

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ p \\ 1 \\ -2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} q \\ 2 \\ 4 \\ 4 \\ -4 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -2 \\ -2 \\ -4 \\ -3 \\ 0 \end{bmatrix},$$

$$w_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad w_2 = \begin{bmatrix} -10 \\ 4 \\ 0 \\ r \\ 1 \end{bmatrix}.$$

Compute dot products:

$$v_1 \cdot w_1 = 1(-1) + 2(-1) + p(1) + 1(0) + (-2)(0) = -1 - 2 + p = p - 3 = 0 \Rightarrow p = 3.$$

$$v_1 \cdot w_2 = 1(-10) + 2(4) + p(0) + 1(r) + (-2)(1) = -10 + 8 + 0 + r - 2 = r - 4 = 0 \Rightarrow r = 4.$$

$$v_2 \cdot w_1 = q(-1) + 2(-1) + 4(1) + 4(0) + (-4)(0) = -q - 2 + 4 = -q + 2 = 0 \Rightarrow q = 2.$$

Check remaining conditions for consistency: $v_2 \cdot w_2 = q(-10) + 2(4) + 4(0) + 4(r) + (-4)(1) = -10q + 8 + 4r - 4 = -10(2) + 4(4) + 4 = -20 + 16 + 4 = 0$. $v_3 \cdot w_1 = (-2)(-1) + (-2)(-1) + (-4)(1) + (-3)(0) + 0(0) = 2 + 2 - 4 = 0$. $v_3 \cdot w_2 = (-2)(-10) + (-2)(4) + (-4)(0) + (-3)(r) + 0(1) = 20 - 8 - 3r = 12 - 3(4) = 0$. All hold.

Thus, the values are

$$\boxed{p = 3, \quad q = 2, \quad r = 4}.$$

Rubric:

- 1 points for recognizing orthogonality of $C(D)$ and $N(D^T)$.
- 3 points for correctly setting up and solving the dot product equations.
- 1 point for the final answer.

3. a) Convert the given vectors ($\vec{v}_1, \vec{v}_2, \vec{v}_3$ and $\vec{v}_4$) into orthonormal vectors using the Gram- Schmidt process. Verify your result by showing that the dot product of any two vectors in the obtained set is zero. 12 (CO3) (PO1)

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \\ 2 \end{bmatrix}, \vec{v}_4 = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

Solution:

Given vectors are:

$$\vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \\ 2 \end{bmatrix}, \quad \vec{v}_4 = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

We apply the Gram-Schmidt process to obtain an orthonormal basis.

Step 1: Orthonormalize $\vec{v}_1$

$$\vec{u}_1 = \vec{v}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix}$$

$$\|\vec{u}_1\| = \sqrt{2^2 + 0^2 + (-1)^2 + 3^2 + (-2)^2} = \sqrt{4 + 0 + 1 + 9 + 4} = \sqrt{18} = 3\sqrt{2}$$

$$\vec{e}_1 = \frac{\vec{u}_1}{\|\vec{u}_1\|} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} \frac{2}{3\sqrt{2}} \\ 0 \\ -\frac{1}{3\sqrt{2}} \\ \frac{3}{3\sqrt{2}} \\ -\frac{2}{3\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{3} \\ 0 \\ -\frac{\sqrt{2}}{6} \\ \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{3} \end{bmatrix}$$

(You may keep as fractions with $\sqrt{2}$ or rationalize later, let's keep simplified the form.)

Step 2: Orthonormalize $\vec{v}_2$

$$\vec{u}_2 = \vec{v}_2 - \text{proj}_{\vec{u}_1} \vec{v}_2 = \vec{v}_2 - \frac{\vec{v}_2 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1$$

$$\vec{v}_2 \cdot \vec{u}_1 = (-2)(2) + (2)(0) + (-1)(-1) + (1)(3) + (0)(-2) = -4 + 0 + 1 + 3 + 0 = 0$$

Since dot product is zero, $\vec{v}_2$ is already orthogonal to $\vec{u}_1$.

$$\vec{u}_2 = \vec{v}_2 = \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\|\vec{u}_2\| = \sqrt{(-2)^2 + 2^2 + (-1)^2 + 1^2 + 0^2} = \sqrt{4 + 4 + 1 + 1 + 0} = \sqrt{10}$$

$$\vec{e}_2 = \frac{\vec{u}_2}{\|\vec{u}_2\|} = \frac{1}{\sqrt{10}} \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{2}{\sqrt{10}} \\ \frac{2}{\sqrt{10}} \\ -\frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \\ 0 \end{bmatrix}$$

Step 3: Orthonormalize $\vec{v}_3$

$$\vec{u}_3 = \vec{v}_3 - \text{proj}_{\vec{u}_1} \vec{v}_3 - \text{proj}_{\vec{u}_2} \vec{v}_3 = \vec{v}_3 - \frac{\vec{v}_3 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1} \vec{u}_1 - \frac{\vec{v}_3 \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2} \vec{u}_2$$

Compute dot products:

$$\vec{v}_3 \cdot \vec{u}_1 = (3)(2) + (1)(0) + (-1)(-1) + (-2)(3) + (2)(-2) = 6 + 0 + 1 - 6 - 4 = -3$$

$$\vec{u}_1 \cdot \vec{u}_1 = 18$$

$$\vec{v}_3 \cdot \vec{u}_2 = (3)(-2) + (1)(2) + (-1)(-1) + (-2)(1) + (2)(0) = -6 + 2 + 1 - 2 + 0 = -5$$

$$\vec{u}_2 \cdot \vec{u}_2 = 10$$

Thus,

$$\vec{u}_3 = \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \\ 2 \end{bmatrix} - \frac{-3}{18}\begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix} - \frac{-5}{10}\begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \\ 2 \end{bmatrix} + \frac{1}{6}\begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$

Compute term by term: First term: $\begin{bmatrix} 3 \\ 1 \\ -1 \\ -2 \\ 2 \end{bmatrix}$ Second: $\frac{1}{6}\begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 0 \\ -1/6 \\ 1/2 \\ -1/3 \end{bmatrix}$ Third: $\frac{1}{2}\begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -1/2 \\ 1/2 \\ 0 \end{bmatrix}$ Sum:

$$\begin{bmatrix} 3 + 1/3 - 1 \\ 1 + 0 + 1 \\ -1 - 1/6 - 1/2 \\ -2 + 1/2 + 1/2 \\ 2 - 1/3 + 0 \end{bmatrix} = \begin{bmatrix} 2 + 1/3 \\ 2 \\ -1 - 2/3 \\ -2 + 1 \\ 2 - 1/3 \end{bmatrix} = \begin{bmatrix} 7/3 \\ 2 \\ -5/3 \\ -1 \\ 5/3 \end{bmatrix}$$

So $\vec{u}_3 = \begin{bmatrix} 7/3 \\ 2 \\ -5/3 \\ -1 \\ 5/3 \end{bmatrix}$. Multiply by 3 to avoid fractions: $\vec{u}_3 = \frac{1}{3}\begin{bmatrix} 7 \\ 6 \\ -5 \\ -3 \\ 5 \end{bmatrix}$. Norm:

$$\|\vec{u}_3\| = \sqrt{(7/3)^2 + 2^2 + (-5/3)^2 + (-1)^2 + (5/3)^2} = \sqrt{\frac{49}{9} + 4 + \frac{25}{9} + 1 + \frac{25}{9}}$$

$$= \sqrt{\frac{49 + 25 + 25}{9} + 5} = \sqrt{\frac{99}{9} + 5} = \sqrt{11 + 5} = \sqrt{16} = 4$$

Thus,

$$\vec{e}_3 = \frac{\vec{u}_3}{\|\vec{u}_3\|} = \frac{1}{4}\begin{bmatrix} 7/3 \\ 2 \\ -5/3 \\ -1 \\ 5/3 \end{bmatrix} = \begin{bmatrix} 7/12 \\ 1/2 \\ -5/12 \\ -1/4 \\ 5/12 \end{bmatrix}$$

Step 4: Orthonormalize $\vec{v}_4$

$$\vec{u}_4 = \vec{v}_4 - \text{proj}_{\vec{u}_1}\vec{v}_4 - \text{proj}_{\vec{u}_2}\vec{v}_4 - \text{proj}_{\vec{u}_3}\vec{v}_4$$

$$= \vec{v}_4 - \frac{\vec{v}_4 \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 - \frac{\vec{v}_4 \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2}\vec{u}_2 - \frac{\vec{v}_4 \cdot \vec{u}_3}{\vec{u}_3 \cdot \vec{u}_3}\vec{u}_3$$

Compute dot products:

$$\vec{v}_4 \cdot \vec{u}_1 = (-2)(2) + (-3)(0) + (1)(-1) + (3)(3) + (-1)(-2) = -4 + 0 - 1 + 9 + 2 = 6$$

$$\vec{v}_4 \cdot \vec{u}_2 = (-2)(-2) + (-3)(2) + (1)(-1) + (3)(1) + (-1)(0) = 4 - 6 - 1 + 3 + 0 = 0$$

$$\vec{v}_4 \cdot \vec{u}_3 = (-2)(7/3) + (-3)(2) + (1)(-5/3) + (3)(-1) + (-1)(5/3) = -\frac{14}{3} - 6 - \frac{5}{3} - 3 - \frac{5}{3}$$

Combine: $-\frac{14}{3} - \frac{5}{3} - \frac{5}{3} = -\frac{24}{3} = -8$; then $-6 - 3 = -9$; total $-8 - 9 = -17$.

$$\vec{u}_3 \cdot \vec{u}_3 = 16 \quad (\text{since } \|\vec{u}_3\| = 4)$$

Thus,

$$\vec{u}_4 = \vec{v}_4 - \frac{6}{18}\vec{u}_1 - \frac{0}{10}\vec{u}_2 - \frac{-17}{16}\vec{u}_3 = \vec{v}_4 - \frac{1}{3}\vec{u}_1 + \frac{17}{16}\vec{u}_3$$

$$\Rightarrow \vec{u}_4 = \begin{bmatrix} -2 \\ -3 \\ 1 \\ 3 \\ -1 \end{bmatrix} - \frac{1}{3}\begin{bmatrix} 2 \\ 0 \\ -1 \\ 3 \\ -2 \end{bmatrix} + \frac{17}{16}\begin{bmatrix} 7/3 \\ 2 \\ -5/3 \\ -1 \\ 5/3 \end{bmatrix} = \begin{bmatrix} -3/16 \\ -7/8 \\ -7/16 \\ 15/16 \\ 23/16 \end{bmatrix}$$

Compute norm:

$$\|\vec{u}_4\| = \frac{1}{16}\sqrt{(-3)^2 + (-14)^2 + (-7)^2 + 15^2 + 23^2} = \frac{1}{16}\sqrt{9 + 196 + 49 + 225 + 529} = \frac{1}{16}\sqrt{1008}$$

$1008 = 16 \cdot 63 = 16 \cdot 9 \cdot 7 = 144 \cdot 7$. So $\sqrt{1008} = 12\sqrt{7}$. Thus $\|\vec{u}_4\| = \frac{12\sqrt{7}}{16} = \frac{3\sqrt{7}}{4}$. Therefore,

$$\vec{e}_4 = \frac{\vec{u}_4}{\|\vec{u}_4\|} = \frac{1}{(3\sqrt{7}/4)} \cdot \frac{1}{16}\begin{bmatrix} -3 \\ -14 \\ -7 \\ 15 \\ 23 \end{bmatrix} = \frac{4}{3\sqrt{7}} \cdot \frac{1}{16}\begin{bmatrix} -3 \\ -14 \\ -7 \\ 15 \\ 23 \end{bmatrix} = \frac{1}{12\sqrt{7}}\begin{bmatrix} -3 \\ -14 \\ -7 \\ 15 \\ 23 \end{bmatrix}$$

We can rationalize or leave as is. For neatness, $\vec{e}_4 = \begin{bmatrix} -\frac{1}{4\sqrt{7}} \\ -\frac{7}{6\sqrt{7}} \\ -\frac{7}{12\sqrt{7}} \\ \frac{5}{4\sqrt{7}} \\ \frac{23}{12\sqrt{7}} \end{bmatrix}$ but we keep the fraction form.

Verification: Show dot product of any two orthonormal vectors is zero (we check a couple):

$$\vec{e}_1 \cdot \vec{e}_2 = \left(\frac{2}{3\sqrt{2}}\right)\left(-\frac{2}{\sqrt{10}}\right) + 0 + \left(-\frac{1}{3\sqrt{2}}\right)\left(-\frac{1}{\sqrt{10}}\right) + \left(\frac{3}{3\sqrt{2}}\right)\left(\frac{1}{\sqrt{10}}\right) + \left(-\frac{2}{3\sqrt{2}}\right)(0)$$

$$= -\frac{4}{3\sqrt{20}} + \frac{1}{3\sqrt{20}} + \frac{3}{3\sqrt{20}} = \frac{-4 + 1 + 3}{3\sqrt{20}} = 0$$

Similarly, others can be checked and will be zero.

The orthonormal vectors are:

$$\vec{e}_1 = \begin{bmatrix} \frac{\sqrt{2}}{3} \\ 0 \\ -\frac{\sqrt{2}}{6} \\ \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{3} \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} -\frac{2}{\sqrt{10}} \\ \frac{2}{\sqrt{10}} \\ -\frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} \\ 0 \end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix} \frac{7}{12} \\ \frac{1}{2} \\ -\frac{5}{12} \\ -\frac{1}{4} \\ \frac{5}{12} \end{bmatrix}, \quad \vec{e}_4 = \frac{1}{12\sqrt{7}} \begin{bmatrix} -3 \\ -14 \\ -7 \\ 15 \\ 23 \end{bmatrix}.$$

Rubric:

- 1 point for correctly identifying first orthonormal vector.
- 2 points for correctly identifying second orthonormal vector.
- 3 points for correctly identifying third orthonormal vector.
- 4 points for correctly identifying fourth orthonormal vector.
- 1 point for showing dot product of any two orthonormal vectors is zero.
- 1 point for correctly normalizing all vectors.

b) Show that if λ is an eigenvalue of an invertible matrix A and $\mathbf{x}$ is a corresponding eigenvector, then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} and $\mathbf{x}$ is a corresponding eigenvector (without using a mathematical example). 6 (CO3) (PO1)

Solution:

Given that A is an invertible matrix, λ is an eigenvalue of A, and $\mathbf{x} \neq \mathbf{0}$ is a corresponding eigenvector. By definition,

$$A\mathbf{x} = \lambda\mathbf{x}.$$

Since A is invertible, $\lambda \neq 0$ (otherwise $\mathbf{x}$ would be in the nullspace of A, contradicting invertibility). Multiply both sides on the left by A^{-1}:

$$A^{-1}(A\mathbf{x}) = A^{-1}(\lambda\mathbf{x})$$
$$\Rightarrow \quad (A^{-1}A)\mathbf{x} = \lambda A^{-1}\mathbf{x}$$
$$\Rightarrow \quad I\mathbf{x} = \lambda A^{-1}\mathbf{x}.$$

Thus,

$$\mathbf{x} = \lambda A^{-1}\mathbf{x}.$$

Dividing both sides by λ (nonzero) yields

$$A^{-1}\mathbf{x} = \frac{1}{\lambda}\mathbf{x}.$$

Hence, $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} and $\mathbf{x}$ is a corresponding eigenvector.

Rubric:

- 1 points for stating the eigenvalue equation and noting $\lambda \neq 0$.
- 4 points for correctly manipulating the equation to obtain $A^{-1}\mathbf{x} = \frac{1}{\lambda}\mathbf{x}$.
- 1 point for the final conclusion.

c) In a 3-dimensional vector space, determine whether an orthogonal complement subspace exists in each of the following cases. If it does not exist, briefly explain why. If it does exist, describe the nature of the orthogonal complement. 3 + 3 (CO3) (PO1)

i. A subspace spanned by three mutually orthogonal vectors in the vector space.

ii. A plane spanned by two linearly independent vectors.

Solution:

In a 3-dimensional vector space ($\mathbb{R}^3$) , every subspace has a unique orthogonal complement. So,

i. A subspace spanned by three mutually orthogonal vectors in the vector space. If the three vectors are nonzero, they form an orthogonal basis of the entire 3 dimensional space. Thus the subspace is the whole space. The orthogonal complement of the whole space is the zero subspace $\{\mathbf{0}\}$. Hence the orthogonal complement exists and is the trivial subspace containing only the zero vector.

ii. A plane spanned by two linearly independent vectors. Two linearly independent vectors span a 2 dimensional subspace (a plane through the origin). In $\mathbb{R}^3$, the orthogonal complement of a plane is **a line through the origin perpendicular to the plane**. This line is 1 dimensional, spanned by a vector orthogonal to both spanning vectors . Thus the orthogonal complement exists and is a straight line through the origin.

Rubric:

- 1 points (times 2) for mentioning whether the orthogonal complement exists or not.
- 2 points (times 2) for correctly identifying what the orthogonal complement subspace would contain.

d) The eigenvalues and eigenvectors of a matrix M are $\lambda_1 = 2$ and $\lambda_2 = 4$ with corresponding eigenvectors $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, respectively. What should be the nature of the matrix M based on the given information? Determine matrix M, and verify whether your assumption is correct. 6 (CO3) (PO1)

Solution:

Given, The eigenvalues and eigenvectors of a matrix M are $\lambda_1 = 2$ and $\lambda_2 = 4$ with corresponding eigenvectors $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, respectively.

Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. By definition of eigenvectors and eigenvalues:

$$Mv_1 = \lambda_1 v_1 \quad \Rightarrow \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = 2\begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

$$Mv_2 = \lambda_2 v_2 \quad \Rightarrow \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 4\begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Performing the matrix multiplications yields the system:

$$\begin{aligned} -a + b &= -2 \qquad (1) \\ -c + d &= 2 \qquad (2) \\ a + b &= 4 \qquad (3) \\ c + d &= 4 \qquad (4) \end{aligned}$$

Solving (1) and (3): adding gives $2b = 2 \Rightarrow b = 1$; then $a = 4 - b = 3$. Solving (2) and (4): adding gives $2d = 6 \Rightarrow d = 3$; then $c = 4 - d = 1$.
Thus,

$$\boxed{M = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}}.$$

The eigenvectors are orthogonal ($v_1 \cdot v_2 = (-1)(1) + (1)(1) = 0$), and for a real matrix with distinct eigenvalues, orthogonal eigenvectors imply the matrix is symmetric. Indeed, M is symmetric because $M^\top = M$, confirming the assumption.

Rubric:

- 2 points for correctly setting up the eigenvalue equations.
- 6 points for solving the system correctly / determining matrix M.
- 2 point for stating the nature (symmetric) and any supporting logic.

4. a) Given, $Ax = b$, for 10 (CO2) (PO1)

$$A = \begin{bmatrix} -2 & 1 & 5 \\ 3 & 2 & 7 \\ -5 & 3 & 9 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

where,

$$det(A_1) = 54, det(A_2) = -189, det(A_3) = 27$$

A_i is the matrix similar to matrix A but its i-th column is replaced by vector b, for example,

$$A_i = \begin{bmatrix} b_1 & 1 & 5 \\ b_2 & 2 & 7 \\ b_3 & 3 & 9 \end{bmatrix}$$

Retrieve vector x and b using your knowledge of Matrices, Determinants and Cramer's rule.

Solution:

Given, $Ax = b$ for

$$A = \begin{bmatrix} -2 & 1 & 5 \\ 3 & 2 & 7 \\ -5 & 3 & 9 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

and the determinants

$$\det(A_1) = 54, \quad \det(A_2) = -189, \quad \det(A_3) = 27,$$

where A_i is A with its i-th column replaced by b.

First, we compute $\det(A)$ using expansion along the first row:

$$\begin{aligned}\det(A) &= (-2)\begin{vmatrix}2 & 7\\ 3 & 9\end{vmatrix} - (1)\begin{vmatrix}3 & 7\\ -5 & 9\end{vmatrix} + (5)\begin{vmatrix}3 & 2\\ -5 & 3\end{vmatrix}\\ &= (-2)(2\cdot 9 - 7\cdot 3) - 1\big(3\cdot 9 - 7\cdot(-5)\big) + 5\big(3\cdot 3 - 2\cdot(-5)\big)\\ &= (-2)(18-21) - 1(27+35) + 5(9+10)\\ &= (-2)(-3) - 1(62) + 5(19) = 6 - 62 + 95 = 39.\end{aligned}$$

So $\det(A) = 39$.
By Cramer's rule,

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{54}{39} = 18/13,$$

$$x_2 = \frac{\det(A_2)}{\det(A)} = \frac{-189}{39} = -63/13,$$

$$x_3 = \frac{\det(A_3)}{\det(A)} = \frac{27}{39} = 9/13.$$

Thus,

$$x = \begin{bmatrix}18/13\\ -63/13\\ 9/13\end{bmatrix}.$$

Now we compute $b = Ax$:

$$b = \begin{bmatrix}-2 & 1 & 5\\ 3 & 2 & 7\\ -5 & 3 & 9\end{bmatrix}\begin{bmatrix}\frac{18}{13}\\ -\frac{63}{13}\\ \frac{9}{13}\end{bmatrix} = \frac{1}{13}\begin{bmatrix}-2\cdot 18 + 1\cdot(-63) + 5\cdot 9\\ 3\cdot 18 + 2\cdot(-63) + 7\cdot 9\\ -5\cdot 18 + 3\cdot(-63) + 9\cdot 9\end{bmatrix}$$

$$= \frac{1}{13}\begin{bmatrix}-36 - 63 + 45\\ 54 - 126 + 63\\ -90 - 189 + 81\end{bmatrix} = \frac{1}{13}\begin{bmatrix}-54\\ -9\\ -198\end{bmatrix} = \begin{bmatrix}-54/13\\ -9/13\\ -198/13\end{bmatrix}.$$

Therefore,

$$b = \begin{bmatrix}-54/13\\ -9/13\\ -198/13\end{bmatrix}.$$

Rubric:

- 1 points for correctly computing $\det(A)$.
- 4 points for correctly determining x via Cramer's rule.
- 4 points for correctly determining $b = Ax$.
- 1 point each for the final vectors x and b (times 2).

b) Given, 10 (CO2) (PO1)

$$B = \begin{bmatrix}3 & -2 & 0 & -4\\ 5 & -6 & 7 & -8\\ -1 & 2 & -3 & 4\\ -5 & 6 & -7 & 8\end{bmatrix}$$

Apply the False Expansion Theorem on matrix B to show that

$$\sum_{i=1}^{n} b_{xi}C_{yi} = 0 \quad (\text{where } x \neq y)$$

While solving, clearly mention each b_{xi} and C_{yi} that you use. Here, x and y represent row numbers, n represents the total number of rows/columns, b_{ij} represents the entry in the i-th row and j-th column of matrix B, and C_{ij} represents the cofactor of entry b_{ij}.)

Solution:

Given,

$$B = \begin{bmatrix} 3 & -2 & 0 & -4 \\ 5 & -6 & 7 & -8 \\ -1 & 2 & -3 & 4 \\ -5 & 6 & -7 & 8 \end{bmatrix}$$

Let us apply the False Expansion Theorem with rows $x = 1$ and $y = 2$ $(x \neq y)$. We need to show

$$\sum_{i=1}^{4} b_{1i}C_{2i} = 0,$$

where b_{1i} are the entries of row 1 and C_{2i} are the cofactors of row 2.
First compute the cofactors C_{2i} for $i = 1, 2, 3, 4$:

$$C_{21} = (-1)^{2+1}M_{21} = -\begin{vmatrix} -2 & 0 & -4 \\ 2 & -3 & 4 \\ 6 & -7 & 8 \end{vmatrix} = -(-24) = 24,$$

$$C_{22} = (-1)^{2+2}M_{22} = \begin{vmatrix} 3 & 0 & -4 \\ -1 & -3 & 4 \\ -5 & -7 & 8 \end{vmatrix} = 44,$$

$$C_{23} = (-1)^{2+3}M_{23} = -\begin{vmatrix} 3 & -2 & -4 \\ -1 & 2 & 4 \\ -5 & 6 & 8 \end{vmatrix} = -(-16) = 16,$$

$$C_{24} = (-1)^{2+4}M_{24} = \begin{vmatrix} 3 & -2 & 0 \\ -1 & 2 & -3 \\ -5 & 6 & -7 \end{vmatrix} = -4.$$

Now form the sum using the entries of row 1:

$$\sum_{i=1}^{4} b_{1i}C_{2i} = b_{11}C_{21} + b_{12}C_{22} + b_{13}C_{23} + b_{14}C_{24}.$$

Substituting the values:

$$= (3)(24) + (-2)(44) + (0)(16) + (-4)(-4) = 72 - 88 + 0 + 16 = 0.$$

Thus, $\sum_{i=1}^{n} b_{xi}C_{yi} = 0$ for $x \neq y$, as required.

Rubric:

- 1 point for correct interpretation of the question/equation.
- 2 points (times 4) for correctly computing the cofactors/minors of each entry.
- 1 point for correct final result (zero).

c) Determine the determinant of matrix C using the Big Formula. 10 (CO2) (PO1)

$$C = \begin{bmatrix} -2 & 0 & 3 & 0 \\ 0 & 7 & 1 & 0 \\ 4 & 0 & 0 & -5 \\ 0 & -9 & 0 & 6 \end{bmatrix}$$

Solution:

Given,

$$C = \begin{bmatrix} -2 & 0 & 3 & 0 \\ 0 & 7 & 1 & 0 \\ 4 & 0 & 0 & -5 \\ 0 & -9 & 0 & 6 \end{bmatrix}$$

Applying the Big Formula,

$$det(C) = \begin{vmatrix} -2 & 0 & 0 & 0 \\ 0 & 7 & 1 & 0 \\ 4 & 0 & 0 & -5 \\ 0 & -9 & 0 & 6 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 3 & 0 \\ 0 & 7 & 1 & 0 \\ 4 & 0 & 0 & -5 \\ 0 & -9 & 0 & 6 \end{vmatrix}$$

$$det(C) = \begin{vmatrix} \boxed{-2} & 0 & 0 & 0 \\ 0 & 7 & \boxed{1} & 0 \\ 4 & 0 & 0 & \boxed{-5} \\ 0 & \boxed{-9} & 0 & 6 \end{vmatrix} + \begin{vmatrix} 0 & 0 & \boxed{3} & 0 \\ 0 & \boxed{7} & 1 & 0 \\ \boxed{4} & 0 & 0 & -5 \\ 0 & -9 & 0 & \boxed{6} \end{vmatrix}$$

$$det(C) = -(2 \times 1 \times 5 \times 9) - (3 \times 7 \times 4 \times 6) = -90 - 504 = -594$$

Rubric:

- 2 points (times 2) for correctly identifying the non-zero combinations.
- 2 points (times 2) for correctly calculating their determinants.
- 2 point for correct answer.

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	12	Least Square Regression
1.b	8	Prediction and Outlier Detection
1.c	10	Better Fit Line
2.a	15	SVD
2.b	10	Diagonalization
2.c	5	Orthogonality of Subspaces
3.a	12	Gram-Schmidt Process
3.b	6	Eigenvalues of Inverse Matrix
3.c	6	Orthogonal Complement
3.d	6	Eigenvalues and Eigenvectors
4.a	10	Cramer's Rule
4.b	10	False Expansion
4.c	10	Big Formula of Determinants
Total	120	